

Technology Stack

Date	01 October 2026
Team ID	SWTID-2026-1380
Project Name	PocketSmartAI: Your Smart Budget & Recommendation Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js with HTML5, CSS3 and JavaScript	Reusable components for the budget form, recommendation cards and plan summary. Responsive design keeps the app simple to use on both phones and desktops.
2	Backend / Server-Side	Python with Flask (REST API)	Quick to build the request handling, recommendation and budget-check logic, and easy to connect to AI services.
3	Database / Data Storage	PostgreSQL	Relational storage for users, planning requests, product and service data and saved plan history, with reliable relationships between them.
4	Cloud / Hosting / Deployment	Render (backend) and Vercel (frontend)	Simple deployment with low-cost, free-tier options suited to a student project, and room to

			scale if usage grows.
5	Version Control & CI/CD	Git and GitHub, with GitHub Actions	Team collaboration and version history, project submission through the GitHub repository, and automated testing on every push.
6	Third-Party APIs / Other Tools	AI / LLM API (e.g., Google Gemini API) and product / service data sources (shopping and event platform APIs)	Enables the personalized recommendations (Process 2) and supplies the options and prices used for budget-checked Home Interior, Party and Jewelry plans.